Homework 1

CFRM 505

Due Sunday, April 12 at 11:59pm

1 Problem 1

Consider the random variables X and Y with the following joint distribution:

$$f_{XY}(x,y) = \begin{cases} \frac{1}{Z}\left(xy^2 + 2xy\right) & 0 < x, y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

where Z is some constant. Calculate the following quantities analytically (**not** by simulation).

1.1 Part a

$A(x,y): 0 \le x \le 0.5,\ 0.5 - x \le y \le 1$

$$\mathbb{P}[X + Y < 0.5].$$

1.2 Part b

$$\mathbb{E}[X].$$

1.3 Part c

$$\mathrm{Var}[Y].$$

1.4 Part d

$$\mathrm{Cov}[X, Y].$$

1.5 Part e

$$\mathrm{Corr}[Y, X].$$

1.6 Part f

$$\mathbb{P}[X > 0.5 \mid Y > 0.5].$$

1.7 Part g

$$\mathbb{E}[X \mid Y = 0.25].$$

One approach is to just sim uniform $(-2, 3)$

$X \sim (-2, 3)$

$X = mU + b$

$X = 5U - 2$

2 Problem 2

Use Monte Carlo integration to estimate

$$\int_{-2}^{3} 3x^2 + 2x \, \mathrm{d}x.$$

(The exact answer is 40. You don't need to rederive it.)

Use a sample size of at least $n = 100,000$.

3 Problem 3

Let $U \sim U(0, 1)$ and $X = 1 - U$. Use Monte Carlo integration to confirm that

$$\mathbb{E}[X] = \frac{1}{2} \text{ and } \mathrm{Var}[X] = \frac{1}{12} \text{ and } \mathrm{Cov}[U, X] = -\frac{1}{12}.$$

Use a sample size of at least $n = 100,000$.

4 Problem 4

Consider the integral

$$\int_{0}^{\infty} xe^{-x} \, \mathrm{d}x.$$

(The exact value of this integral is 1. You don't need to rederive it.)

4.1 Part a

Use the u-substitution $u = x/(1 + x)$ to show that this integral is equivalent to the expectation

$$\mathbb{E}\left[\frac{U}{(1-U)^3} e^{-U/(1-U)}\right]$$

where $U \sim U(0, 1)$. Estimate this expected value using Monte Carlo integration with a sample size of at least $n = 100,000$.

4.2 Part b

Use the u-substitution $u = 1 - e^{-x}$ to show that this integral is equivalent to the expectation

$$\mathbb{E}\left[-\ln(1 - U)\right]$$

where $U \sim U(0, 1)$. Estimate this expected value using Monte Carlo integration with a sample size of at least $n = 100,000$.

4.3 Part c

Show that this integral is equivalent to the expectation

$$\mathbb{E}[X]$$

where $X \sim \text{Exp}(1)$. Estimate this expected value using Monte Carlo integration with a sample size of at least $n = 100,000$. You should generate samples of X directly using the code `np.random.exponential(1, size=n)`.

5 Problem 5

Let U_1 and U_2 be i.i.d. uniform random variables on $(0, 1)$. Define $X = \max\{U_1, U_2\}$.

5.1 Part a

Find $\theta = \mathbb{P}[X \leq 0.5]$ by hand (not by simulation).

5.2 Part b

We can use Monte Carlo integration to estimate θ by generating n i.i.d. samples of U_1 with `u1 = np.random.uniform(0, 1, size=n)` and another n i.i.d. samples of U_2 with `u2 = np.random.uniform(0, 1, size=n)`, then using these to generate n samples of X.

Use this approach to estimate θ with a sample size of at least $n = 100,000$.

5.3 Part c

If $U_1 \sim U(0, 1)$, then we know that $1 - U_1$ is also uniformly distributed on $(0, 1)$. We might therefore be tempted to make our code more efficient by generating n i.i.d. samples of U_1 with `u1 = np.random.uniform(0, 1, size=n)` and then generating U_2 with `u2 = 1 - u1`, then using these two to generate n samples of X as before.

Explain why this approach will **not** effectively estimate θ. What goes wrong? What answer will you get for θ?

Derive the following by hand and then use Monte Carlo simulation to estimate them. (In both cases, use a sample size of at least $n = 100,000$.)